MILITARY
SPECIFICATION

MIL-D-11554A(SigC)
17 November 1953
Superseding
MIL-D-11554(SigC)
23 November 1951

DYNAMOTOR-POWER SUPPLY DY-88( )/GRC-9

1. SCOPE

1.1 Scope.-This specification covers the requirements for one type of multi-input dynamotor power supply to supply all the necessary voltages and currents for operation of Receiver-Transmitter RT-77( )/GRC-9 of Radio Set AN/GRC-9( ). (See 6.4.)

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-D-24	Dynamotors.
MIL-V-95	Vibrators; Interrupter and Self-Rectifying.
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed (For Weight of Contents Not in Excess of 1000 Pounds).
MIL-B-107	Boxes, Wood, Wirebound (Overseas Type).
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-P-116	Preservation, Methods of.
JAN-P-117	Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-B-131	Barrier-Material; Water-Vaporproof, Flexible.
MIL-S-901	Shock-Proof Equipment, Class HI (High-Impact): and Tests for (Shipboard Application).
MIL-P-11268	Parts, Materials, and Processes Used in Signal Corps Equipment: General Specification.
MIL-C-13077	Cable, Special Purpose, Electrical (Cordage CO-212 and Cable, Special Purpose, Electrical WD-32/U).

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment.

DRAWINGS

SIGNAL CORPS

460-14	Group B-4 Sampling Plans.
SC-D-15914	Moisture-Resistance Test Cycle for Ground Signal Equipment.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Procurement model. -The procurement model shall be understood as the one which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, the equipment shall conform to the following:

3.1.1 Construction. -Construction of the equipment shall conform to the procurement model. In case of conflict between the specification and the model, the specification shall govern.

3.1.2 Functions. -The equipment shall incorporate all functions provided by the procurement model.

3.1.3 Performance. -Performance of the equipment shall be at least equal to that of the procurement model. When any performance characteristics of the procurement model is not specified herein, the following shall apply: As soon as the procurement model is received by the contractor, he shall adjust it for optimum performance and prepare to conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance to enable his representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

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3.2 Preproduction samples.-

3.2.1 Quantities.-After award of contract, in addition to the samples listed in Specification MIL-P-11268, two complete equipments shall be furnished for preproduction inspection and approval.

3.2.2 Compliance with specified requirements.-

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.5), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 Test data.-The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.2.4 Preproduction inspection.-Preproduction samples may be subjected by the Government to group A inspection and nondestructive group B and group C inspection specified in tables II, III, and IV, respectively, and to the preproduction inspection specified in Table I. Unlisted nondestructive inspection on production samples of equipment and unlisted destructive or nondestructive inspection on separate preproduction samples of parts or materials may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order; (1) vibration, (2) bounce, (3) shock, (4) immersion; and (5) life; other preproduction inspection may precede, follow or be interspersed between the foregoing.

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TABLE I. -Preproduction inspection

Inspection	Requirement Paragraph	Test Paragraph
Vibration	3.11.1	4.12
Bounce	3.11.2	4.13
Shock	3.11.3	4.14
Immersion	3.7.2	4.15
Moisture-resistance	3.4	4.16
Life	3.9	4.17

3.2.5 Reference standards. -After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Parts, materials, and processes; general. -In additional to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.4 Service conditions. -The equipment shall be constructed to withstand long periods of storage (both before and after unpacking), and continuous operation under any combination of service conditions listed below, after which it shall meet specified performance. The following criteria are considered the minimum requirements to insure satisfactory performance of the equipment in field service:

a. Extreme temperature. - from -40° F. to +125° F. (operating)
from -80° F. to +160° F. (storage)

b. Relative humidity. - As high as 100 percent at specified temperature including moisture due to temperature changes

c. Vibration, bounce and shock. - (See 3.11.)

3.5 Circuit label. -A circuit diagram (schematic, wiring, and cabling) shall be furnished and placed within the equipment. The diagrams shall be in the form of labels meeting the applicable requirements of Specification MIL-P-11268 and shall be printed on white book paper laminated between two sheets of minimum 0.015-inch, clear, transparent, polyvinyl-chloride-acetate sheets. The two sheets shall be bonded together to seal against moisture. The label shall be capable of withstanding extreme service conditions.

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3.6 Construction. - The equipment shall be a complete, compact assembly capable of operation from a 6-, 12-, or 24-volt vehicular electrical system to provide the specified output voltages necessary to operate Receiver-Transmitter RT-77( )/GRC-9. Two separate power circuits shall be provided in the unit; a dynamotor supply to furnish power for the regulated transmitter filament and high voltage for the transmitter plate, and a vibrator power supply to supply plate power to the receiver under "Send" and "Standby" conditions.

3.7 Mechanical requirements. -

3.7.1 Dimensions and weight. - The dimensions and weight of the equipment furnished on order shall not exceed that of the model.

3.7.2 Immersion-proofing. - The equipment shall be constructed to meet the test specified in 4.15.

3.8 Electrical requirements. -

3.8.1 Input. - The equipment shall operate over the following input voltage ranges, when operated with any of the specified loads and temperature requirements and with Receiver-Transmitter RT-77/GRC-9

6-volt position - - - - - - - - - - 5.5 to 8 volts d-c inclusive
12-volt position - - - - - - - - - - 11 to 16 volts d-c inclusive
24-volt position - - - - - - - - - - 22 to 32 volts d-c inclusive

3.8.2 Output (full load conditions). - With an input of 7.2, 14.5, or 29 volts, the equipment shall deliver the following outputs, when connected to the resistive loads listed below (equivalent to the average power requirements of the various operating conditions of Receiver-Transmitter RT-77( )/GRC-9):

Transmitter-Receiver Vibrator Unit. -

	Rated Output Voltage (d-c)	Output Current (d-c) (ma)	Maximum Output Voltage (d-c)	Minimum Output Voltage (d-c)	Equivalent Load Resistance (ohms)
Transmitter plate	580	100	630	530	5800
Receiver plate	120	45	140	105	2625
Receiver filament	1.5	500	1.6	1.4	3.0
Transmitter Filament	6.5	2000	6.8	6.2	3.25
Keying relay	6.9	575	7.4	6.4	12

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3.8.3 Output (standby conditions).-With an input of 7.2, 14.5, or 29 volts, the equipment shall deliver the following outputs, when connected to the resistive loads listed below (equivalent to that of Receiver-Transmitter RT-77 ( )/GRC-9 operated in the standby position):

Receiver Standby Vibrator Unit.-

	Rated Output Voltage (d-c)	Output Current (d-c) (ma)	Maximum Output Voltage (d-c)	Minimum Output Voltage (d-c)	Equivalent Load Resistance (ohms)
Receiver plate	150	25.0	170	130	5900
Receiver filament	1.6	500	1.8	1.4	3.0

3.8.4 Regulation.-The maximum voltage regulation of the transmitter plate supply circuit with an input voltage of 7.2, 14.5, or 29.0 volts, shall not exceed 15 percent when measured from average load to minimum load as specified in 4.13. Regulation of the receiver or receiver standby plate supply circuit shall be such as to obtain satisfactory operation of a regulator tube JAN-OC3/VR105. (Tube shall not be contained in the power supply). The receiver and transmitter filament voltages shall be within the minimum and maximum voltage requirements of 3.8.2 with the power unit operating at any input voltage within the ranges specified in 3.8.1. A protective device should be incorporated in the equipment, if necessary, to prevent the transmitter filament voltage from exceeding a value of 8 volts, under any condition of filament lead.

3.8.5 Ripple.-The ripple voltage in the output circuits of the equipment shall not exceed the following values when measured in accordance with 4.7, and when the power supply is subjected to the temperature tests of 4.10:

Circuit	Ripple voltage (average V)
Transmitter plate	4.0
Receiver plate	0.1
Receiver filament	0.005
Transmitter filament	0.05
Keying relay	0.3
Receiver plate (standby)	0.1

3.8.6 Efficiency.-The efficiency of the equipment shall be not less than 37 percent at an input of 7.2 volts; 40 percent at an input of 14.5 volts; and 40 percent at an input of 29 volts when calculated under the conditions specified in 4.8.

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3.8.7 Interference filtering.-Adequate means of filtering shall be incorporated in the equipment to prevent radiated and conducted interference with radio communication equipment operating in the frequency range of 10 kc to 20 kc inclusive, and whose power source is the same vehicular electrical system as the Dynamotor-Power Supply DY-88( )/GRC-9. The filtering shall also prevent interference with all radio equipment powered by Dynamotor-Power Supply DY-88( )/GRC-9 when the former is operating in the frequency range of 2.0 mc to 12.0 mc, inclusive.

3.8.7.1 The signal plus noise to noise ratio of the radio receiver shall not be less than 10 db when tested in accordance with 4.6.

3.8.7.2 With the equipment furnishing power to Receiver-Transmitter RT-77( )/GRC-9, the transmitter cw, mcw or voice modulated signal shall show no evidence of ripple, warble, hash, or distortion when tested as specified in 4.6.1.

3.8.8 Power Cable (Cable Assembly, Special Purpose, Electrical CX-2031( )/U).-The power cable used to connect the equipment to the vehicular terminal shall be Cable Assembly, Special Purpose, Electrical, CX-2031( )/U. The cord shall be eight feet long using Cordage CO-212 in accordance with Specification MIL-C-13077. The cord shall have an Amphenol plug, Catalogue No. 164-201-1P(.520), or equal on one end, and 8" pigtails with 3/8" terminal lugs and approved type covering for the ground wire on the other end.

3.8.9 Output terminal.-The output circuits of the equipment shall terminate in Socket SO-141 which shall be as per model.

3.8.10 Protective devices.-Suitable fuses or circuit breakers shall be provided in the equipment to protect the equipment from damage due to overloads.

3.8.11 Vibrator.-The vibrator shall be type VB024S01 in accordance with Specification MIL-V-95. The vibrator shall be capable of withstanding the life test of 4.17 when installed in Dynamotor-Power Supply DY-88( )/GRC-9.

3.8.12 Dynamotor DY-97( )/GRC-9.-The dynamotor shall be a multiple-input, single output, continuous duty, d-c machine capable of operation from a 6-, 12-, or 24-volt source and shall conform to the applicable requirements of Specification MIL-D-24 and Specification Sheet MIL-D-24/12 thereto. The dynamotor shall have 5 leads as follows:

Low Voltage	High Voltage
+6 Volts	+580 volts
+12 volts	
+24 volts	
negative for 6, 12, 24 volts	

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3.9 Life.-The equipment including major components shall be capable of satisfactory operation, without permanent degradation, for 1000 hours under the conditions specified in 4.17 at any ambient temperature up to plus 125° F.

3.10 Heat rise.-With the equipment having reached thermal stability, at any specified test input voltage, under the duty cycle specified in 4.17, the temperature rise of any component shall not exceed its maximum rated value when tested at any ambient temperature up to +125° F.

3.11 Vibration, bounce, and shock.-

3.11.1 Vibration.-The equipment shall have no mechanical resonance below 55 cycles per second when shock mounts are blocked or removed. There shall be no evidence of physical damage resulting from the test of 4.12.

3.11.2 Bounce.-The equipment shall be subjected to the bounce test of 4.13. After completion of this test, the equipment shall be capable of full specification performance and shall not have suffered any physical damage except surface abrasion from the bounce test.

3.11.3 Shock test.-The equipment shall be subjected to the shock test of 4.14 and shall meet full specified performance after completion of this test. No physical damage shall occur as a result of this test.

3.12 Bounce conditioning.-After final assembly, each equipment on contract, with shock mounts removed or blocked, shall be placed in its normal operating position of the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this conditioning for 1 minute and then shall be capable of meeting the inspection of section 4.

3.13 Technical literature, tools, and running spare parts.-Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.14 Interchangeability.-Corresponding equipments and replaceable subassemblies and parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are to be considered replaceable shall be as determined by the contracting officer.

3.15 Workmanship.-All parts, components, and subassemblies of Dynamotor-Power Supply DY-88( )/GRC-9 shall be manufactured, finished, and assembled in a thoroughly workmanlike manner.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general.-Government inspection shall consist of group A, group B, and group C inspection as specified in 4.1.1 through 4.1.3.2. (See 6.3.1.) Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met the group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.1.1 Group A inspection.-Group A inspection shall be in accordance with table II and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE II.-Government group A inspection

Inspection or Test	Requirement Paragraph	Test Paragraph	AQL Major	AQL Minor
Visual and Mechanical		4.2	2.5 dphu	10.0 dphu
Electrical				
Input	3.8.1	4.3	)	
Output	3.8.2	4.4	)	
Regulation	3.8.4	4.5	) 2.5 dphu	
Ripple Voltage	3.8.5	4.7	) for the	
Efficiency	3.8.6	4.8	) Group	
Noise Measurement	3.8.7	4.6	)	
Air Seal Test	3.7.2	4.9	1.0%	

NOTE: Prior to acceptance inspection, the equipment shall meet the requirements of 3.12.

4.1.2 Group B inspection.-Group B inspection shall be in accordance with table III.

4.1.2.1 Order of inspection within group B.-Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

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4.1.2.2 Sampling procedure.-Samples shall be selected in an unbiased manner by the Government inspector. Unless otherwise specified, inspection shall be started using level II of the specified drawing. If the number of defective sample units from a lot exceeds the acceptance number specified on the drawing, the lot shall be rejected. Disposition of rejected product (sample units and lots) shall be in accordance with Standard MIL-STD-105 and 4.1.5.

4.1.2.3 Procedure in case of failure.-When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

TABLE III.-Government group B inspection

Inspection or Test	Requirement Paragraph	Test Paragraph	Sampling Plan
Vibration Bounce Shock	3.11.1 3.11.2 3.11.3	4.12 4.13 4.14	DWG. 460-14 For Group

4.1.3 Group C inspection.-This shall consist of the group C inspection specified in table IV.

4.1.3.1 Selection of samples.-Two equipments for each group C inspection shall be selected each month in an unbiased manner by the Government inspector.

4.1.3.2 Noncompliance.-If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action to be taken on the process and affected portion of product. If the Government inspector does not consider that the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE IV.-Government group C inspection

Inspection or Test	Requirement Paragraph	Test Paragraph	Sampling Plan
Temperature Heat Rise Life	3.4 3.10 3.9	4.10 4.11 4.17	Two samples per month (see paragraph 4.1.3.1)

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4.1.4 Reinspection of conforming group B and group C sample units.-Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection (after repair of all visible damage).

4.1.5 Disposition of nonconforming product.-When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.2 Mechanical and visual inspection.-Parts and components shall be inspected for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, finishes, etc. shall be examined for workmanship. Dimensions shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Visual and mechanical defects shall be classified by the Government as major and minor in accordance with the definitions of Standard MIL-STD-105.

4.3 Input.-The equipment shall be tested for compliance with the requirements of 3.8.1.

4.4 Output.-The output voltages of the equipment, when measured with standard resistive loads equal to average operating conditions of the radio ser, shall be within the limits specified in 3.8.2 and 3.8.3. Measurements shall be made with a high resistance voltmeter having an accuracy of 1 percent or better.

4.5 Regulation.-The voltage regulation of the transmitter plate circuit shall be determined by operating the equipment with the specified d-c input voltages and with resistive loads equal to average operating conditions of the radio set, as specified in 3.8.2. The regulation, specified in 3.8.4, shall then be calculated from the following formula:

$$\text{Percent regulation} = \frac{(E_1 - E_2)100}{E_2}$$

where E_1 = Voltage across the transmitter plate output circuit with the 5800-ohm resistor disconnected,
and E_2 = Voltage across the transmitter plate output circuit with the 5800-ohm resistor connected as the load.

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4.6 Noise measurement. -The equipment shall be connected to the radio set and operated with the 7.2, 14.5 or 29 volts d-c input. The "R-F Gain Control" on the radio set shall be adjusted to maximum position. A signal generator shall be connected to the antenna terminal of the radio set and adjusted for 10 microvolts output, modulated 30 percent, 400 cycles. An output meter adjusted to the output impedance of the radio set shall be connected to the phone jack. The "AF Gain Control" shall be adjusted to obtain 10 milliwatts output; the modulation shall then be removed and the output level noted. The signal-plus-noise to noise ratio shall not be less than 10 db when measured at the following frequencies with the radio set adjusted for "Standby" operation and repeated for "Send High Power" operation with the key or microphone switch open. (See 3.8.7.)

BAND	FREQUENCY (Megacycles)
1	6.6 and 12
2	3.6 and 6.6
3	2.0 and 3.6

4.6.1 A dummy load, consisting of a 0-1 ampere R.F. ammeter in series with a 20-ohm noninductive resistor and a 70-micromicrofarad capacitor shall be connected between the antenna binding post and the metal housing of Receiver-Transmitter RT-77( )/GRC-9. The transmitted frequency shall be adjusted to 4.8 megacycles. The antenna selector switch, and the antenna tuning control, shall be adjusted for highest antenna current. A second Receiver-Transmitter RT-77( )/GRC-9 shall be tuned to the transmitter signal, and the signal listened to; there shall be no evidence of warble, ripple, hash, or distortion. This test shall be conducted with the transmitter adjusted for "high" power operation on "CW", "MCW", and "PHONE" positions. (See 3.8.7.2).

4.7 Ripple voltage test. -The ripple voltage in the output circuits shall be measured with an electronic voltmeter, Ballantine Model 300, or equal, and shall not exceed the limits specified in 3.8.5 when the equipment is operated with approximately 29 volts d-c input, and when connected to the resistive loads specified in 3.8.2 and 3.8.3.

4.8 Efficiency. -Efficiency of the power unit shall be calculated from data obtained in 3.8.2 (Output (full load conditions)), using the following formulas:

$$\text{Percent efficiency} = \frac{\text{Total power Output} \times 100}{\text{Total power input}}$$

$$\text{Power (Watts)} = \frac{E^2 \text{ (volts)}}{R \text{ (ohms)}}$$

Total power output shall be calculated by adding the following:

$$\frac{(\text{Transmitter plate voltage})^2}{5800 \text{ ohms}} = ?$$

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$$\frac{(\text{Receiver plate voltage})^2}{2625 \text{ ohms}} = ?$$

$$\frac{(\text{Transmitter filament voltage})^2}{3.25 \text{ ohms}} = ?$$

$$\frac{(\text{Receiver filament voltage})^2}{3 \text{ ohms}} = ?$$

$$\frac{(\text{Keying relay voltage})^2}{12 \text{ ohms}} = ?$$

Total power output = ?

4.9 Air-seal test. - The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.10 Temperature tests. -

4.10.1 Heat test. - The equipment shall be placed in a heat chamber for three days at an ambient temperature of +160° F. The temperature shall then be decreased to +125° F. and the equipment allowed to reach thermal equilibrium. The unit shall then be operated for a period of six hours, at duty cycle specified in 4.17, with rated output loads as specified in 3.8.2 and 3.8.3 and at an ambient temperature of +125° F. Leads connected to the output of the equipment shall extend from the heat chamber, and measurements of output shall be made at regular intervals throughout the test. The output shall not vary more than 15 percent from those taken at the beginning of the test.

4.10.2 Cold test. - The equipment shall be placed in a cold chamber for 3 days at an ambient temperature of -80° F. The temperature shall then be raised to -40° F. and the equipment allowed to reach thermal equilibrium. The unit shall then be operated for a period of six hours, at the duty cycle specified in 4.17 with rated output loads as specified in 3.8.2 and 3.8.3, and at the ambient temperature of -40° F. Leads connected to the output of the equipment shall extend from the cold chamber, and measurements of output shall be made at regular intervals throughout the test. The output shall not vary more than 15 percent from those taken at the beginning of the test.

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4.11 Heat rise. - The equipment shall be tested for compliance with the requirements of 3.10. The method of temperature measurement shall be submitted to the contracting officer for approval.

4.12 Vibration test. - The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.11.1 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.12(d)).

(b) The equipment shall be vibrated successively for 15 minutes in each of three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of sub-assemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Mass., or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indicators.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or inaccessibility of its sub-assemblies or component parts, the unit shall be broken down into logical sub-assemblies for the above vibration test. Such sub-assemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.13 Bounce test. - The equipment, on its shock mounts shall be secured to the Vehicular Adapter Plate and placed on the table of the Package Tester, both as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The plate shall be constrained from horizontal motion of more than two inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90 degrees, in the same direction, at the end of each 3/4-hour period.

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4.14 Shock test. - The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" as shown in Specification MIL-S-901. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate.

4.15 Immersion test. - The equipment, with cover properly clamped in place, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be approximately 40° F. above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water at approximately its initial temperature, or the temperature of the water shall be maintained constant by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.16 Moisture-resistance test for equipment. -

4.16.1 Test conditions. -

(a) Do not move equipments from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the components. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed components with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure.

4.16.2 Test procedure. - The equipments will be tested as follows:

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Condition at 77° ± 5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.16.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC-D-15914. The measurements specified in 4.16.3 shall be made during the periods shown on the drawing and shall comply with 4.16.3.

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(e) After cycling has been completed, condition the equipments for 24 hours at 77° ± 5° F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipments. No repair or replacement of parts shall be made. After adjustment, the equipments shall meet full specified performance for those measurements specified in 4.16.3.

4.16.3 Performance. -The voltage output of the equipment shall be the same as the output voltages as measured before the moisture-resistance test.

4.16.4 Failure. -If any equipment fails to meet performance specified in 4.16.3 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.16.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements as specified in 4.16.2, it does not pass the test.

4.17 Life. -The equipment shall be operable at any ambient temperature up to + 55° C, with rated output loads as specified in 3.8.2 and 3.8.3, from any input voltage of the ranges specified in 3.8.1 using the following duty cycle:

(a) 5 minutes transmit - - - - - - - full load.

(b) 10 minutes transmit - - - - - - no load.

(c) 15 minutes standby (receive) after a total of 45 minutes of the above cycle (a and b).

(d) Every 24 hours, during a standby (receive) period, the equipment will be shut off for a period not to exceed 5 minutes. The equipment shall then be tested for starting at the minimum voltage of the particular voltage range (see 3.8.1) at which the life test is being conducted.

During, and after completion of the life test, the equipment shall be capable of meeting the requirements of 3.8.2.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment (immediate use).-Dynamotor-Power Supply DY-88( )/GRC-9 shall be packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 For Military overseas or domestic shipment involving extended storage.-Preserve and package Dynamotor-Power Supply DY-88( )/GRC-9 in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs:

5.1.2.1 Technical manual.-Package each technical manual Method IC-3 as follows: Inclose each technical manual within a close-fitting Type II Class b bag, conforming to Specification JAN-P-117. Properly seal closure.

5.1.2.2 Dynamotor-Power Supply DY-88( )/GRC and bags of hardware.-Package each dynamotor Method IC-5 as follows: Operating fuses shall remain within the unit. Place nine fuses within the spare fuse compartment on the face of the dynamotor. The remaining fuses shall be placed within the spare parts box on the chassis of the dynamotor. Four sets of brushes shall be in use and installed within the unit.

5.1.2.3 Spare brushes.-Package each set of four sets of brushes individually Method IA-8 as follows: Inclose set of brushes within a close-fitting water-vapor-proof bag fabricated of barrier material qualified for Class 2 application in accordance with the requirements of Specification MIL-D-131. Heat seal closure, and place with the spare parts box.

5.1.2.4 Operating vibrator and ballast tubes shall remain in unit. Spare vibrator and ballast tubes shall be installed in spare parts socket located on the chassis of dynamotor, no packaging necessary.

5.1.2.5 Electrical spare parts items and accessories.-Electrical spare parts items and accessories determined in quantity and number from the bid request or contract shall not require packaging. Stow within the spare parts box attached to the chassis of dynamotor.

5.1.2.6 Dynamotor-Power Supply DY-88( )/GRC-9.-Pad the left and right sides of the dynamotor with cells of rigid commercial corrugated fiberboard (200 pound Mullen Test). The pads shall be fabricated in such a manner so as to build up the sides of the dynamotor to the dimensions of the projecting shock mounts. Pad the face of the dynamotor with a cell of rigid commercial corrugated fiberboard (275 pound Mullen Test). The pads shall be fabricated so as to extend 1/2 inch beyond the fasteners on the face of the unit. Place the bagged hardware kits within the cells on side of unit. Pad the top of the dynamotor with a cell of rigid commercial corrugated (275 pound Mullen Test) in such a manner so as to build up the top of the case to the height of the face flange.

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5.1.2.7 Cable CX-2031/U and Cord CD-1086.-Wind each item into a coil of suitable dimensions and tie at 3 places with cotton tape securing the coils with the ties, and place within a commercial corrugated fiberboard carton (200 pound Mullen Test). The width and height of the carton is to be of size to fit snugly within the w-board carton containing dynamotor.

5.1.2.8 Place each cushioned dynamotor and packaged cable together within a close-fitting corrugated fiberboard box conforming to Specification JAN-P-108 for W-5c 275 pound Mullen Test box. The carton containing cable packaged as specified in 5.1.2.7 shall be placed against back face of cushioned dynamotor. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape conforming to Specification JAN-P-127.

5.2 Packing.-

5.2.1 For domestic shipment (immediate use).-Dynamotor-Power Supply DY-88( )/GRC-9 shall be packed so as to insure arrival at destination in satisfactory condition and in containers complying with the Consolidated Freight Rules, at lowest rates.

5.2.2 For Military overseas or domestic shipment involving extended storage.-Each dynamotor-Power Supply DY-88( )/GRC-9 shall be packed in a wood cleated plywood, nailed wood or wirebound wood box conforming to the requirements of Specifications JAN-P-105, JAN-P-106 and MIL-B-107 respectively. Fabricate the boxes to fit contents snugly. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment.-Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

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6. NOTES

6.1 Intended use.-Dynamotor-Power Supply DY-88( )/GRC-9 is intended to be used to operate Receiver-Transmitter RT-77( )/GRC-9 of Radio Set AN/GRC-9.

6.2 Ordering data.-Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required.

(c) Samples required (see 3.2).

(d) Marking and shipping of samples.

(e) Facsimiles required.

(f) Technical literature required (see 3.13).

(g) Quantity of tools and running spare parts required (see 3.13).

6.3 Contractual requirements.-It is recommended that the following be included in the contract:

6.3.1 Additional inspection.-It should be understood that additional examination, measurement, and nondestructive testing of equipment may be performed by the Government when considered necessary to determine compliance with this specification and other applicable documents. Accordingly, the Government inspector may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection performed either at a Government laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples.-When items for which production samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should be made by the prime contractor and should include a statement indicating that: the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order and will be at the same plant; and current production is in strict compliance with specified requirements for the new equipment except for such deviations as may be authorized on the current order. The deviations should be described in the statement. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

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6.4 Nomenclature.-The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: DY-88W/GRC-9. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information.

6.5 Location of air-seal test.-It is desirable that the air-seal test (4.9) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all acceptable sample units) be resampled at this location for the purpose of performing this inspection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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